INCH-POUND
MIL-P-15737C(SH)
29 April 1991

SUPERSEDING
MIL-P-15737B(SH)
27 April 1960

MILITARY SPECIFICATION

PADDLES, BOAT, SOLID AND LAMINATED WOOD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid wood paddles and laminated wood paddles for boats and life floats.

1.2 Classification. Paddles should be of the following types, styles, and sizes, as specified (see 6.2):

Type I – Laminated wood construction
- Style A – Standard blade
  - Size 48 – 48 inch length
  - Size 60 – 60 inch length
  - Size 66 – 66 inch length
  - Size 72 – 72 inch length
- Style B – Wide blade
  - Size 48 – 48 inch length
  - Size 60 – 60 inch length
  - Size 66 – 66 inch length
  - Size 72 – 72 inch length

Type II – Solid wood construction
- Style A – Standard blade
  - Size 48 – 48 inch length
  - Size 60 – 60 inch length
  - Size 66 – 66 inch length
  - Size 72 – 72 inch length

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116	Preservation, Method of
MIL-T-704	Treatment and Painting of Material
MIL-W-18142	Wood Preservation Solutions, Oil-Soluble, Ship and Boat Use
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-A-22397	Adhesive, Phenol and Resorcinol Resin Base (For Marine Service Use)

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 523	Standard Test Method for Specular Gloss; (DOD adopted)
D 822	Standard Practice for Operating Light- and Water-Exposure Apparatus (Carbon-Arc Type) for Testing Paint and Related Coatings and Materials; (DOD adopted)
D 2016	Standard Test Methods for Moisture Content of Wood; (DOD adopted)
D 3951	Standard Practice for Commercial Packaging; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance in 4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the currentversion before use.

3.2 Description. Paddles shall be of the shape, dimensions, and construction specified herein.

3.3 Materials.

3.3.1 Wood. Unless otherwise specified 3.3.1.1, 3.3.1.2, or 3.3.1.3 for specific parts, wood used in the construction of laminated and solid wood paddles shall have no splits or openings in the surface of the wood over 1/32 inch deep, and shall contain no wane or bark, rot, dote, incipient decay, wormholes, shake, pith, organic stain caused by various species of fungi, or other strength-reducing defects. Natural color of wood, regardless of variability, is not considered a defect nor is mineral streak. The moisture content of wood used in the paddles shall be not less than 6 nor more than 14 percent. However, wood parts for any given paddle shall not vary more than 4 percent in moisture content at time of fabrication.

3.3.1.1 Type I handle and grip. The handle shall be made of white oak, red oak, birch, hard maple, or of the following species known as "white ash": white ash, green ash, blue ash, and Biltmore ash. Woods known commercially as "pumpkin ash," "swell-butted," or "brash" material shall not be used. The shaft of the handle and the handle grip shall be made of the same species of wood. Cross grain measured in any direction in the lower 2/3 of the handle and shaft (including the portion attached to the blade) shall not exceed a slope of 1 in 15 inches of length. Cross grain measured in any direction in the upper 1/3 portion of the handle shaft shall not exceed a slope of 1 in 12 inches.

3.3.1.2 Type I blade. The blade shall be of yellow poplar, spruce, or basswood, sound, smoothly cut, and well manufactured. Knots and burls less than 3.8 inch in diameter and widely scattered pinworm holes may be permitted in the blade. Cross grain not in excess of a slope of 1 in 12 inches may also be permitted.

3.3.1.3 Type II paddles. Solid wood paddles shall be one of the "white ash" species permitted for handles as specified in 3.3.1.1 and shall be made from solid wood throughout, except for an insert at the tip of the blade as described in 3.4.2 and shown in figure 1. Cross grain shall not exceed 1 in 15 inches of length in any direction except in the upper one-third of the handle, wherein cross grain shall not exceed 1 in 12 inches in length in any direction.

3.3.2 Adhesives. The adhesives used for all gluing operations shall conform to MIL-A-22397. Only adhesives which have been approved for listing on the Qualified Products List shall be used in the fabrication of paddles.

3.4 Construction. Construction shall be as described herein and as shown on figures 2 and 3 for type I paddles or as shown on figure 1 for type II paddles.

3.4.1 Type I paddle. The paddle shall be constructed in accordance with 3.4.1.1 and 3.4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

3.4.1.1 Type I paddle: handle and shaft. The type I paddle handle grip and shaft may be either of solid or parallel laminated construction. The handle grip shall be glued to the shaft.

3.4.1.2 Type I paddle: blade. The blade shall be of laminated three-ply construction. The two outside plies shall have the direction of grain parallel to the length of the paddle and may be one piece or may be glued for width. The center ply or crossbanding shall be approximately 1/24 to 1/16 inch in thickness with the grain placed at right angles to the grain of the outside plies and may be one piece or may be glued for width.

3.4.2 Type II paddle. The solid paddle grip, shaft, and blade shall be of one-piece construction with the grain parallel to the length except that a cross-ply wood insert 1/24 to 1/16 inch in thickness shall be placed with its grain at right angles to the grain of the blade at the mid thickness and in the plane of the tip of the blade, the cross-ply to extend not less than 3 inches into the blade from the tip (see figure 1). The cross-ply insert shall be of one of the species listed in 3.3.1.2 or "white ash".

3.4.3 Preparation for assembly. Preparation for assembly shall be in accordance with 3.4.3.1 and 3.4.3.2.

3.4.3.1 Preparation of type I blade and shaft for assembly. The shaft shall be prepared for assembly by drilling a 5/32-inch hole at the extreme depth of the proposed notch and machining the notch. Machining of the notch shall preferably be done with a bandsaw with at least 1/2-inch blade, operating with a fixture which will ensure uniform taper and straight, smooth surfaces on the sides of the notch. The blade of the paddle shall be grooved on both faces symmetrically about the centerline of the blade, preferably with a flat cutting router operated with a fixture which will ensure uniform taper and smooth flat surfaces in the bottom of the grooves. The dimensions of the notch in the shaft and the grooves in the blade shall be adjusted to each other so that when the two parts are assembled, the mating faces will fit closely enough that the blade will advance to the full depth of the notch.

3.4.3.2 Preparation of type II paddle blade and insert for assembly. A saw kerf 3 inches deep sufficiently wide to obtain a close fit with the insert and adhesive in place shall be cut parallel to and at the mid-thickness of the plane of the blade.

3.4.4 Assembly. Assembly shall be in accordance with 3.4.4.1 and 3.4.4.2.

3.4.4.1 Type I paddles. The shaft shall be bonded to the blade by using an adhesive as specified in 3.3.2. Adhesive spreading, assembly period, pressure application and curing shall be controlled in accordance with the adhesive manufacturer's instructions. The adhesive bond shall pass all required tests (see 4.6.3.1.2 and 4.6.3.3).

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the currentversion before use.

3.4.4.2 Type II paddles. The blade insert shall be bonded to the blade using an adhesive specified in 3.3.2. Adhesive spreading, assembly period, pressure application, and curing shall be controlled in accordance with the adhesive manufacturer's recommendations. The adhesive bonds shall pass all required tests (see 4.6.3.1.2 and 4.6.3.3).

3.5 Shape and dimensions of paddles. Paddles shall conform to the shape and dimensions shown on figures 1 and 3. Except when specifically noted on figures 1 and 3, tolerances for the various paddle length, paddle width, and paddle thickness dimensions shall be as shown in table I.

TABLE I. *Dimension tolerances.*

Dimension	Maximum tolerance for individual measurement (inch) (plus or minus)
Thickness:	
I, T-1, T-2, T-3, and T-4 (see figure 3)	1/32
Other than I, T-1, T-2, T-3, and T-4 (see figure 3)	1/16
Width:	
Less than 1-3/8 inch	1/16
1-3/8 inch to and including 3 inches	1/8
Over 3 inches	3/16
Length:[1]	
B, C, and L-1 on figure 3, and total length	1/4
Radii:	
Greater than 3/8 inch but less than 1 inch	1/8
Greater than 1 inch but less than 15 inches	1/4
15 inches or greater	1

[1]Other length dimension tolerances shall be plus or minus 1/16 inch.

3.5.1 Handle shape. The handle shaft shall be round but a perfect circle is not required. The handle shall be of the specified thickness and width. The difference between thickness and width shall not exceed 1/16 inch. One or more sides of the handle shaft may have flat zones not exceeding 1/4 inch in width after shaping. After sanding, the transition from the flat zones to the curved zones shall be smooth.

3.6 Straightness. Maximum deviation of the centerline of the paddle from a straight line drawn from the center of the end of the handle to the center of the tip of the blade measured at the mid-blade of the paddle shall be as shown in table II.

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.

TABLE II. *Straightness.*

Type of Deviation	Maximum deviation for an individual paddle (inch)
Looking flatwise at the paddle:	
size 48 paddle	3/16
size 60 paddle	3/16
size 66 paddle	1/4
size 72 paddle	1/4
Looking edgewise at the blade:	
size 48 paddle	3/16
size 60 paddle	1/4
size 66 paddle	5/16
size 72 paddle	3/8

3.7 Freedom from cupping. Deviation of the centerline of the paddle blade from a straight line extending from edge to edge at the midpoint of the blade shall not be greater than to cause one side of the blade to assume a flat surface.

3.8 Bending strength. Paddles loaded at the center of the span with the applicable proof loads as shown in table III shall not exceed the maximum deflection nor the maximum permanent set as shown in table III (see 4.6.2).

TABLE III. *Bending strength.*

Paddle size	Proof load (pounds)		Maximum deflection (inches)		Maximum permanent set (inch)
	Style A	Style B	Style A	Style B	
48	129	129	1.5	1.5	0.1
60	100	118	2.5	2.4	0.1
66	90	106	3.1	2.9	0.1
72	96	96	3.5	3.5	0.1

3.9 Resistance of adhesive bonds to delamination. The maximum delamination in various adhesive lines shall not exceed that shown in table IV (see 4.6.3).

3.10 Unpainted paddle surface. Prior to application of coatings, paddles shall be hand or machine dressed, sanded evenly, and smoothly finished with all edges and corners rounded.

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the currentversion before use.

TABLE IV. *Resistance of adhesive bonds to delamination.*

Type of adhesive line	Maximum average percent delamination for all paddles in sample	Maximum percent delamination for any individual paddle in sample
Laminated paddles		
Between laminations of shaft	5	10
Between handle grip and shaft	5	10
Between plies of blade	5	10
Between blade and shaft	15	25
Solid wood paddles		
Between insert blade	5	10

3.11 Coatings. Except for the top 4 ± 1/2 inches, paddles shall be coated in accordance with MIL-T-704, type B, except for color. Unless otherwise specified (see 6.2), the color shall match color number 36076 of FED-STD-595 for Navy purchases and color number 36118 of FED-STD-595 for other purchases. In addition, the dry finish on the coated paddles shall meet the requirements of 3.11.1, 3.11.2, and 3.11.3.

3.11.1 Resistance of paint coating to cyclic exposure. No paint failure shall occur when tested in accordance with the cyclic exposure tests specified in 4.6.3.

3.11.2 Resistance of the paint coating to accelerated weathering. No paint failure shall occur as a result of the accelerated weathering test specified in 4.6.6.

3.11.3 Specular gloss finish on coated paddles. The gloss reading both before and after the accelerated weathering test shall not exceed 8.0 when tested in accordance with the specular gloss test specified in 4.6.5.

3.12 Wood preservation. After sanding to final size and finish, paddles shall be given a 3-minute dip in wood preservative, MIL-W-18142, type C. Adequate time shall be allowed after dip to permit evaporation of solvent. A very light post-dip sanding may be used to assure proper paint adhesion.

3.13 Identification marking. Each paddle shall be marked on the blade near the end of the handle in accordance with MIL-STD-130. Paint or ink used for marking shall be as durable as the paddle coating system and shall provide a definite contrast with the finish coat. Letters shall be not less than 1/8 inch nor more than 5/16 inch in height and shall be clearly legible. Marking shall include the following, as applicable:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

a. Nomenclature (Paddle, boat, laminated wood; or Paddle, boat, wood)
b. Manufacturer code
c. Length (48, 60, 66, or 72)
d. Stock number
e. Specification (MIL-P-15737)
f. Contract number
g. US

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations of 4.5 and the tests of 4.6 (see 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations of 4.5 and the test of 4.6.

4.4.1 Sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

4.4.1.1 Inspection lot. Paddles of the same size, not exceeding 1000, presented at one time for inspection shall be considered a lot for purposes of sampling and inspection.

4.4.1.2 Sampling for examination and tests. Sampling for examination and test shall be in accordance with 4.4.1.2.1 through 4.4.1.2.5. Sampling shall be on an unbiased random basis.

4.4.1.2.1 Sampling for examination of unpainted paddles for materials, construction, quality of surface, separation at adhesive line, and freedom from cupping. Sampling of paddles shall be in accordance with inspection level II of MIL-STD-105.

4.4.1.2.2 Sampling for examination of paddles for shape, dimensions, and straightness. Sampling of paddles shall be in accordance with inspection level S-4 of MIL-STD-105.

4.4.1.2.3 Sampling for visual examination of paint finish and identification marking. Sampling for visual examination of paint finish shall be in accordance with inspection level I of MIL-STD-105.

4.4.1.2.4 Sampling for the bending test. Sampling of paddles for the bending test shall be on a one-hundred percent basis; paddles in the lot shall be tested for bending strength in accordance with 4.6.2.

4.4.1.2.5 Sampling for test of paint finish and adhesive delamination. The sampling for the test of paint finish and adhesive delamination shall be in accordance with 4.4.1.2.5.1 or 4.4.1.2.5.2, as applicable.

4.4.1.2.5.1 Sampling for cyclical exposure test. If the paint finish is to be tested by the cyclical exposure test of 4.6.3 (see 3.11.1), one sample shall be selected in accordance with inspection level S-2 of MIL-STD-105. The required number of paddles shall be randomly selected from the paddles passing the bend test. Paint finish and adhesive delamination shall be tested on the same sample.

4.4.1.2.5.2 Sampling for accelerated weathering test. If the paint finish is to be tested by the accelerated weathering test of 4.6.6, a sample shall be selected for paint finish tests in accordance with inspection level S-2 of MIL-STD-105. A separate sample shall be selected for a test of adhesive delamination in accordance with 4.6.3. The required number of paddles for each sample shall be randomly selected from paddles passing the bend test.

4.4.1.2.5.2 Sampling for accelerated weathering test. If the paint finish is to be tested by the accelerated weathering test of 4.6.6, a sample shall be selected for paint finish tests in accordance with inspection level S-2 of MIL-STD-105. A separate sample shall be selected for a test of adhesive delamination in accordance with 4.6.3. The required number of paddles for each sample shall be randomly selected from paddles passing the bend test.

4.5 Examinations.

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

4.5.1 Examination of unpainted paddles for material, construction, quality of surface, separation at adhesive line, and freedom from cupping. Unpainted paddles selected from a lot shall be examined for quality of surface, separation at adhesive lines, and freedom from cupping.

4.5.2 Examination of paddles for shape, dimensions, and straightness. Sample paddles shall be examined for conformance to the shape, dimension, and straightness requirements of 3.5 and 3.6. In measuring the dimension tolerance, the thickness and width of the handle shaft for each paddle specimen shall be measured at one point. One length dimension, one other width dimension, one other thickness dimensions, and one radii dimension shall also be measured. The same dimensional measurements shall be made for all paddles in the sample.

4.5.3 Examination for paint finish. The contractor shall have certification from the suppliers of the paint coating that the coating meets applicable specification requirements. It shall be the contractor's responsibility to ensure conformance to the applicable paint specification. The paddle manufacturer shall establish and maintain procedures to examine effectively the paint finishing system and the painted paddles to determine conformance to the coverage and color requirements of 3.11.

4.6 Test methods.

4.6.1 Moisture content of wood. The moisture content of the wood shall be determined in accordance with ASTM D 2016.

4.6.2 Static bending strength tests. Bending strength tests shall conducted as specified below to determine conformance to the requirements specified in 3.8. A paddle shall be rated as "reject" if visible damage occurs, if the difference between the gauge reading before and after loading exceeds 0.1 inch, or if the deflection during loading exceeds the maximum deflection values shown in table III. Paddles shall not show visible damage or exceed the foregoing limits for permanent deflection or deflection during loading.

4.6.2.1 Apparatus. The apparatus for testing paddles in static bending shall provide the following attributes:

a. Support the paddle as a simple beam 3 inches from each end.
b. Be able to measure the amount of deflection at the center of the span before and during loading and after the load is removed. The deflection measuring gauge shall be readable to 0.05 inch difference before and after loading.
c. Be able to apply the required load (plus or minus 1.5 pounds) at the load point as shown in table V. The rate of load application shall not exceed 1 inch deflection per second.
d. Loading head shall conform to figure 4. Support points shall be contoured to a 1 to 2 inch radius for the surfaces that contact the specimen during the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

TABLE V. *Bending test loading points.*

Paddle length (inches)	Span (inches)	Point of loading (measured from end of handle) (inches)
48	42	24
60	54	30
66	60	33
72	66	36

4.6.2.2 Test procedure. The tests shall be conducted as follows: Place the paddle in the bending apparatus and set the deflection gauge in contact with the paddle handle at the loading position. Read the deflection gauge indication. Apply the appropriate load at the load point and read the gauge. Then remove the load and adjust the deflection gauge to contact the paddle handle. Read the deflection gauge. Determine the difference between the first and the full load and the first and the final deflection gauge readings.

4.6.3 Cyclic exposure tests.

4.6.3.1 Preparation of test specimens. Specimens for evaluating various features of the paddle by the cyclic exposure test shall be prepared as specified in 4.6.3.1.1 through 4.6.3.1.3.

4.6.3.1.1 Preparation of specimens for testing adhesive lines in handle grip of laminated paddle. The specimen used in testing glue bonds in the handle grip shall be prepared by cutting the upper 6 inches of the handle from the shaft of painted paddles. The paint extending onto this handle section shall be removed by sanding. The sawed surface shall also be sanded smooth.

4.6.3.1.2 Preparation of specimens for testing adhesive bonds in blade and adhesive bonds between shaft and blade of the laminated paddle. The shaft shall be removed from the paddle by cross-cutting at the top of the notch between the shaft and the blade. The blade and remaining attached portion of the shaft shall be bisected by sawing vertically through the thickness of the blade so that the tapered glue lines will be exposed. The sawed surfaces shall be sanded smooth. The curved edge of each bisected section of the blade shall be sanded smooth to expose the glue lines between the blade plies.

4.6.3.1.3 Preparation of specimens for testing adhesive bonds of the insert at the tip of the blade of solid wood blade. The end of the blade shall be sawed off 1-1/2 inches from the tip at a right angle from the long direction of the paddle. The sawed area and the curved end portion of the blade shall be sanded smooth to expose the glue line on all edges.

4.6.3.2 Test procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

4.6.3.2.1 Vacuum pressure cycle. The specimens specified shall be placed in an autoclave and weighted down. Water at 65 to 80 degrees Fahrenheit (°F) shall be admitted into the autoclave in a quantity sufficient to submerge the specimens completely. Thereafter, a vacuum of 20 to 25 inches mercury (Hg) shall be drawn and held for 2 hours. The vacuum shall then be broken, and a pressure of 80 ± 5 pounds per square inch (lb/in^2) shall be applied to the submerged specimens and held for 2 hours. The pressure shall then be released and a vacuum of 20 to 25 inches of Hg shall again be drawn and held for 2 hours while the specimens remain submerged. The vacuum shall be released again and a pressure of 80 ± 5 lb/in^2 shall be applied to the submerged specimens and held for 2 hours.

4.6.3.2.2 Drying period. The specimens shall be dried for 88 hours (3-2/3 days) in air at 80 to 85 °F and 25 to 30 percent relative humidity, and circulating at the rate of 500 ± 50 feet per minute. During the drying period, the specimens shall be placed not less than 2 inches apart with the end surfaces parallel to the stream of air.

4.6.3.2.3 Duration of test. The cyclical exposure test shall consist of three cycles of soaking and drying specified in 4.6.3.2.1 and 4.6.3.2.2 comprising of a total test period of 12 days.

4.6.3.3 Measurement of delamination. Delamination is a term used to express separation of wood surfaces at the glue lines. When the separation is in the wood, even though very close to the glue line, it is termed wood failure or checking. Magnification is often necessary to determine whether the failure is in the glue line or wood. A feeler gauge 0.004 inch thick is convenient for probing into the glue line to determine if separation actually exists.

4.6.3.3.1 Glue lines between blade and shaft. Delamination of glue joints between blade and shaft shall be measured on the full length of the tapered adhesive lines exposed in bisecting the blade.

4.6.3.3.2 Glue lines between plies of the blade. Delamination of glue lines between plies of the blade shall be measured on all exposed adhesive lines including those exposed on the curved edges of the blade and those exposed by bisecting the blade.

4.6.3.3.3 Glue lines at the handle grips. Delamination of glue joints at the handle grip shall be measured on all exposed adhesive lines. If laminated shafts are used, delamination in the shaft shall be measured and reported separately.

4.6.3.3.4 Glue lines between the insert and the blade tip of the solid wood paddle. Delamination of glue lines between the blade and the insert shall be measured on all exposed adhesive lines including those on the curved edges of the blade.

4.6.3.4 Calculations for percent delamination. The percent delamination of a particular type of adhesive line in a test specimen shall be calculated as follows:

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the currentversion before use.

$$\text{Percent delamination} = \frac{\text{Total delamination measured}}{\text{Total length of exposed adhesive line}} \times 100$$

Average percent delamination for a particular type of adhesive line for all specimens shall be calculated as follows:

$$\text{Average percent delamination} = \frac{\text{Sum of percent delaminations for individual specimens}}{\text{Number of specimens}}$$

4.6.4 Observation for paint failure. Each specimen section of the blade which was subjected to the 12-day cyclic exposure test shall be examined for paint failure.

4.6.5 Specular gloss test. Before and after coated paddle blades are subjected to the accelerated weathering test specified in 4.6.6, the coating shall be subjected to the specular gloss test in accordance with ASTM D 523. Prior to conducting the specular gloss test on paddles previously subjected to the accelerated weathering test, the hazy surface deposit shall be removed with a damp cloth and the surface shall be permitted to dry.

4.6.6 Accelerated weathering test. The shaft shall be removed from the paddle by cross-cutting 22-1/2 inches from the tip of the paddle blade. The 22-1/2 inch paddle blade section shall be subjected to the accelerated weathering test in accordance with ASTM D 822 for 100 hours. At the end of the test period paddles shall be inspected for paint failure.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy shipboard stowage fire-retardant requirements.

a. *Treated lumber and plywood.* Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

Levels A and B	–	Type II – weather resistant Category 1 – general use
Level C	–	Type I – non-weather resistant Category 1 – general use.

b. *Fiberboard.* Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

c. *Cushioning and wrapping materials.* The use of excelsior, newspaper, shredded paper (all types), similar hygroscopic or nonneutral materials and all types of loose fill materials for packaging applications such as cushioning, fill, stuffing, and dunnage is prohibited. Materials selected for cushioning and wrapping shall properties (characteristics) for resistance to fire (see 6.7). Cushioning or wrapping materials, as applicable, shall be provided to prevent item and package damage and to prevent free movement of the container contents.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Paddles in quantities of 10 shall be preserved method III in accordance with MIL-P-116, except that the paddles shall be separated from each other with cellulosic cushioning material and placed in unit packs.

5.2.1.1 Unit pack. Unit packs (unit containers) shall conform to MIL-STD-2073-1, appendix F, table I. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option. Containers shall be of type weather resistant and closure in accordance with the appendix of the box specification, with method V applicable to fiberboard boxes.

5.2.2 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be of minimum weight and cube, similar construction, and of uniform size.

5.3.2 Level A, B, and C containers. Material preserved as specified (see 5.2) shall be packed in close-fitting nailed wood or wood-cleated plywood boxes for the level of packing specified (see 6.2), in accordance with MIL-STD-2073-1, appendix C, table VII. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the currentversion before use.

5.3.2.1.1 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that class weather resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding, and non-weather resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.2 Weight. Wood, plywood, and cleated type containers shall not exceed 200 pounds gross weight.

5.3.2.1.3 Waterproofing. Unless otherwise specified (see 6.2), level A and when specified (see 6.2), level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.3 Commercial. Material preserved as specified (see 6.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of 3- or 4-inch nominal wood skids laid flat, or a skid- or sill-type base that will support the material and facilitate handling by mechanical handling equipment during shipment.

5.4 Marking.

5.4.1 Level A, B, C, and commercial. In addition to any special marking required (see 6.2), level A, B, and C interior packs and shipping containers shall be marked in accordance with MIL-STD-2073-1, appendix F, and commercial interior packs and shipping containers shall be marked in accordance with ASTM D 3951. In addition, bar coding shall be applied in accordance with ASTM D 3951. In addition, bar coding shall be applied in accordance with the marking requirements of MIL-STD-2073-1.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Paddles are intended for use with boats and life floats.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Type, style, and size of paddles (see 1.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

d. Whether first article is required (see 3.1)

e. Color, if other than as specified (see 3.11)

f. Fire-retardant material requirements (see 5.1.1)

g. Level of preservation (see 5.2)

h. Container selection (see 5.2.1.1 and 5.3.2)

i. Level of packing (see 5.3)

j. Waterproofing requirements (see 5.3.2.1.3)

k. Special marking (see 5.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
4.1.1	DI-R-4803	Inspection System Program Plan	–
4.3	DI-T-4902	First Article Inspection Report	–

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the currentversion before use.

tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Quality acceptance limits. For guidance in judging acceptable quality, the following limits are suggested.

6.5.1 Visual examination. When inspected according to 4.5.1, individual sample paddles with defects should be rejected. Acceptance or rejection of the lot should be based on an acceptable quality level (AQL) of 1.5 percent defective for paddle lots with excess cupping and separation at adhesive lines, and an AQL of 2.5 percent defective for quality of surface. Failure of any paddle due to improper construction or material should result in rejection of the lot.

6.5.2 Visual examination for shape, dimensions, and straightness. When inspected according 4.5.2, if either the average dimension tolerance requirement of 3.5 or the average straightness deviation requirements of 3.6 is not-met, the entire lot should be rejected. For the maximum individual tolerance requirements of 3.5, acceptance or rejection of the lot should be based on the following levels:

a. Thickness and width of handle shaft: AQL of 1.5 percent defective
b. Width (except for handle shaft): AQL of 4.0 percent defective
c. Thickness (except for handle shaft): AQL of 4.0 percent defective
d. Length: AQL of 4.0 percent defective
e. Radii: AQL of 4.0 percent defective.

For the maximum individual tolerance requirements of 3.6, acceptance or rejection of the lot should be based on an AQL of 2.5 percent defective.

6.5.3 Examination of paint finish. When inspected according to 4.5.3, any paddles in the sample not meeting the finish requirements of 3.11 should be rejected. Acceptance or rejection of the lot represented by the sample should be based on an AQL of 4.0 percent defective.

6.6 Paddle type. Type I and II paddles provide equivalent strength and durability. The type I paddle is slightly lower in weight because of its blade material, and the type II paddle is lower in cost because of its simpler construction. It is recommended that acquisition documents specify "type I or type II" unless there is a valid reason for restricting the order to one type only.

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.

6.7 **Cushioning and wrapping materials (see 5.1.1(c)).** Materials having properties for resistance to fire and acceptable for use within unit packs and shipping containers for Navy shipboard stowage furniture item acquisitions are:

Material	Specification
Paper, kraft treated (fire-resistant)	A-A-1894
Paper, kraft, wrapping	UU-P-268, type II, grade C or D
Fiberboard	PPP-F-320, class-domestic/fire retardant
Plastic film, flexible, cellular	PPP-C-795, class 3 – fire-retardant
Polystyrene expanded, resilient	PPP-C-850, grade SE
Plastic, open cell, cushioning	PPP-C-1842, type I, style B
Bound fiber	PPP-C-1120, type III or IV, grade 1
Rubber, latex foam	MIL-R-5001, grade A
Rubber, Cellular	MIL-R-6130, grade A
Fibrous glass	MIL-C-17435
Polystyrene foam	MIL-P-19644, type II
Rubber, cellular, synthetic	MIl-R-20092, class 5
Polyurethane foam	MIL-P-26514
Polyurethane foam, flexible, open cell	MIL-F-81334
Foam, combustion retardant for cushioning supply items aboard Navy ships	MIL-F-87090

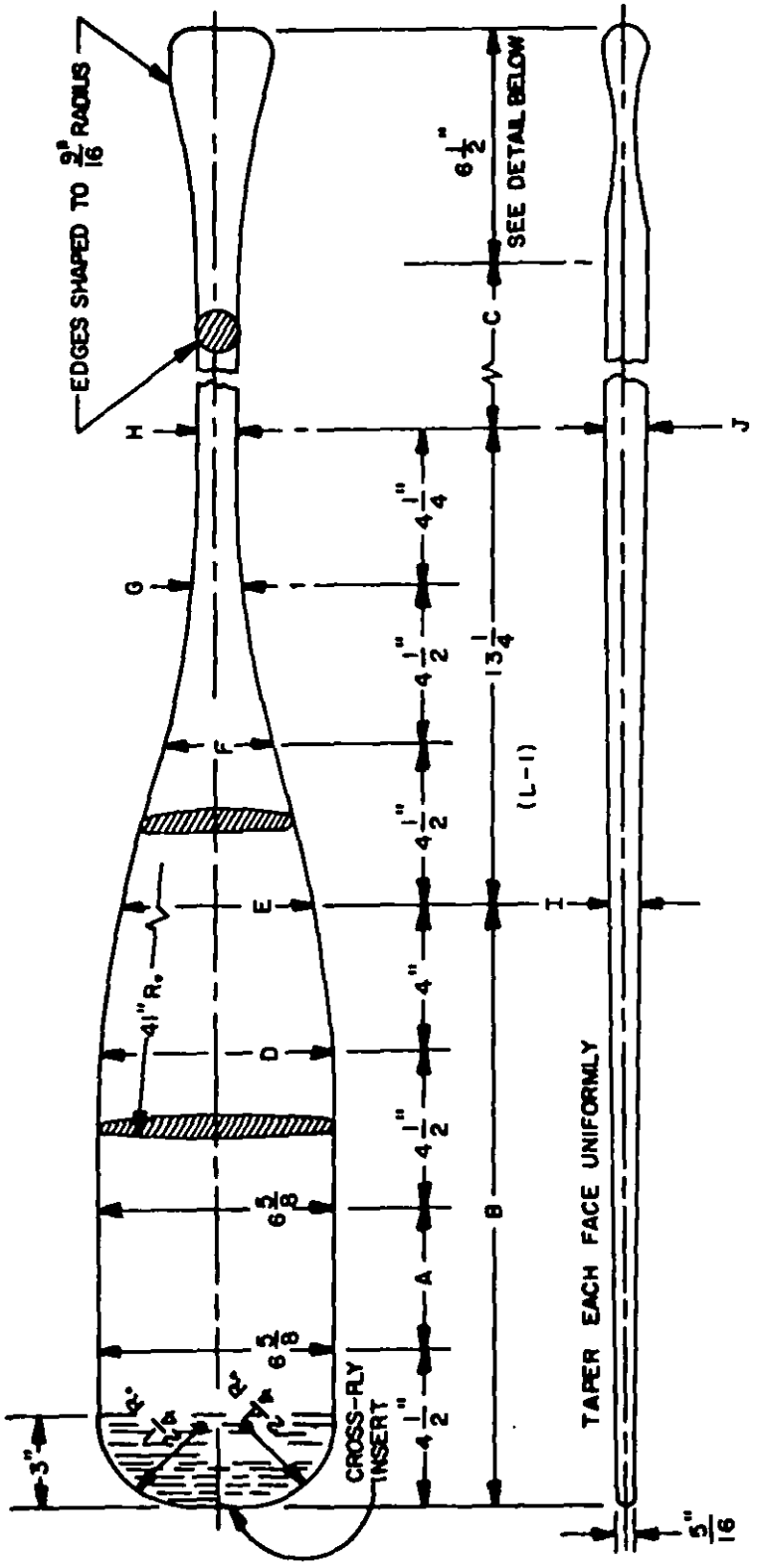
6.8 **Subject term (key word) listing.**

Boat paddle
Life float paddle

6.9 **Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 2040-N177)

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the currentversion before use.



	A	B	C	D	E	F	G	H	I	J	K	M
SIZE 48	0	13	15-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	25/32	1-1/8	1-1/8	1-1/8
SIZE 60	4	17	23-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	25/32	1-1/8	1-1/8	1-1/8
SIZE 66	4	17	29-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	25/32	1-1/8	1-1/8	1-1/8
SIZE 72	4	17	35-1/4	6-5/8	5-7/8	4-3/8	2-3/8	1-3/8	25/32	1-3/8	1-3/16	1-3/16

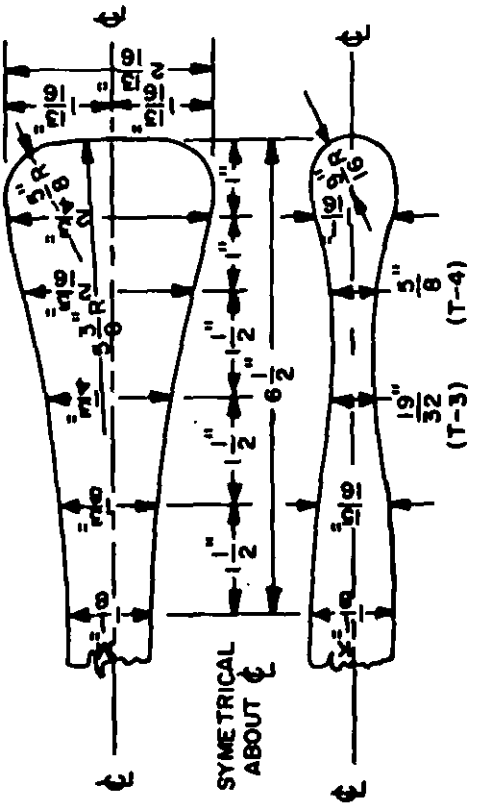


FIGURE 1. *Shape and dimensions of type II paddle.*

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.

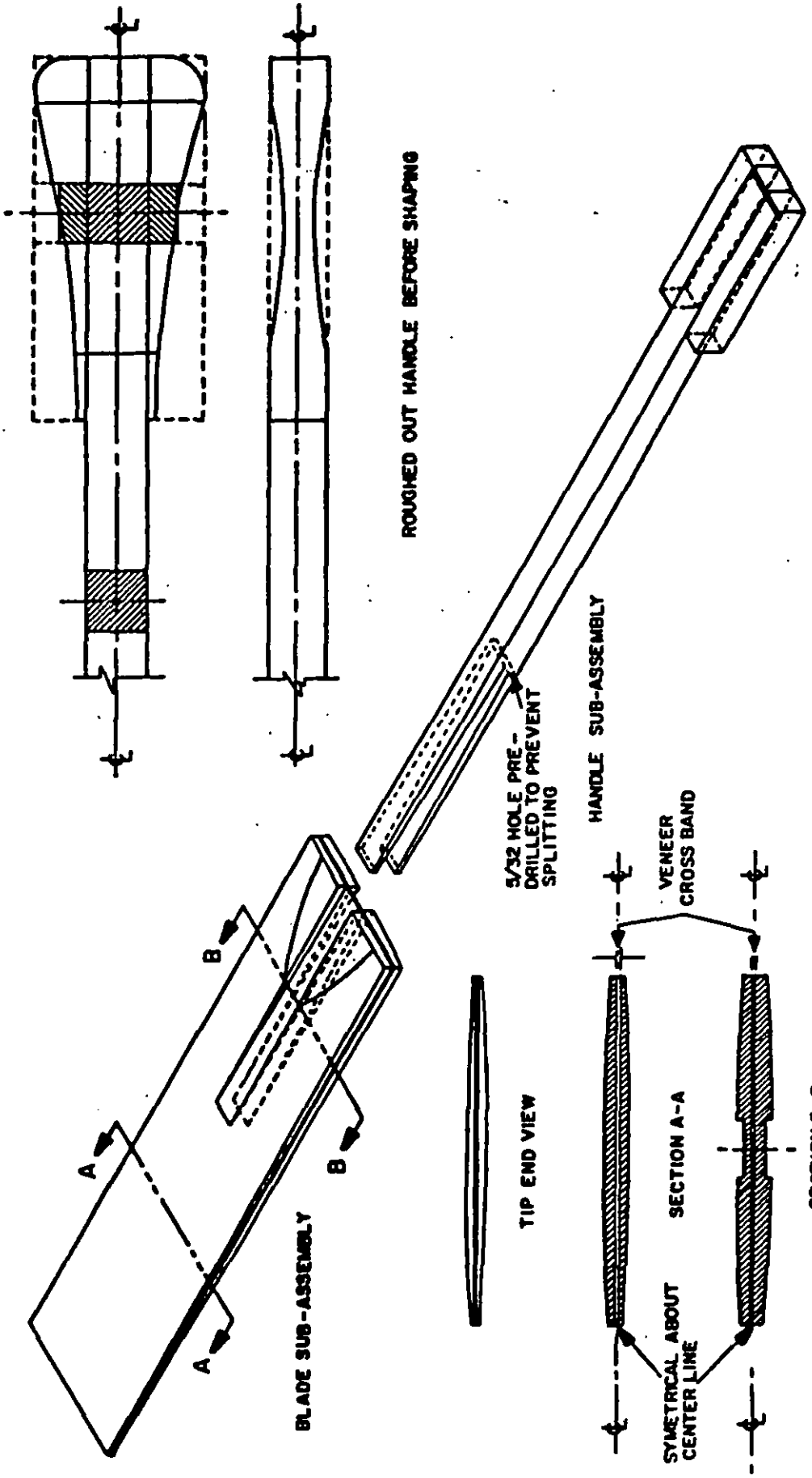
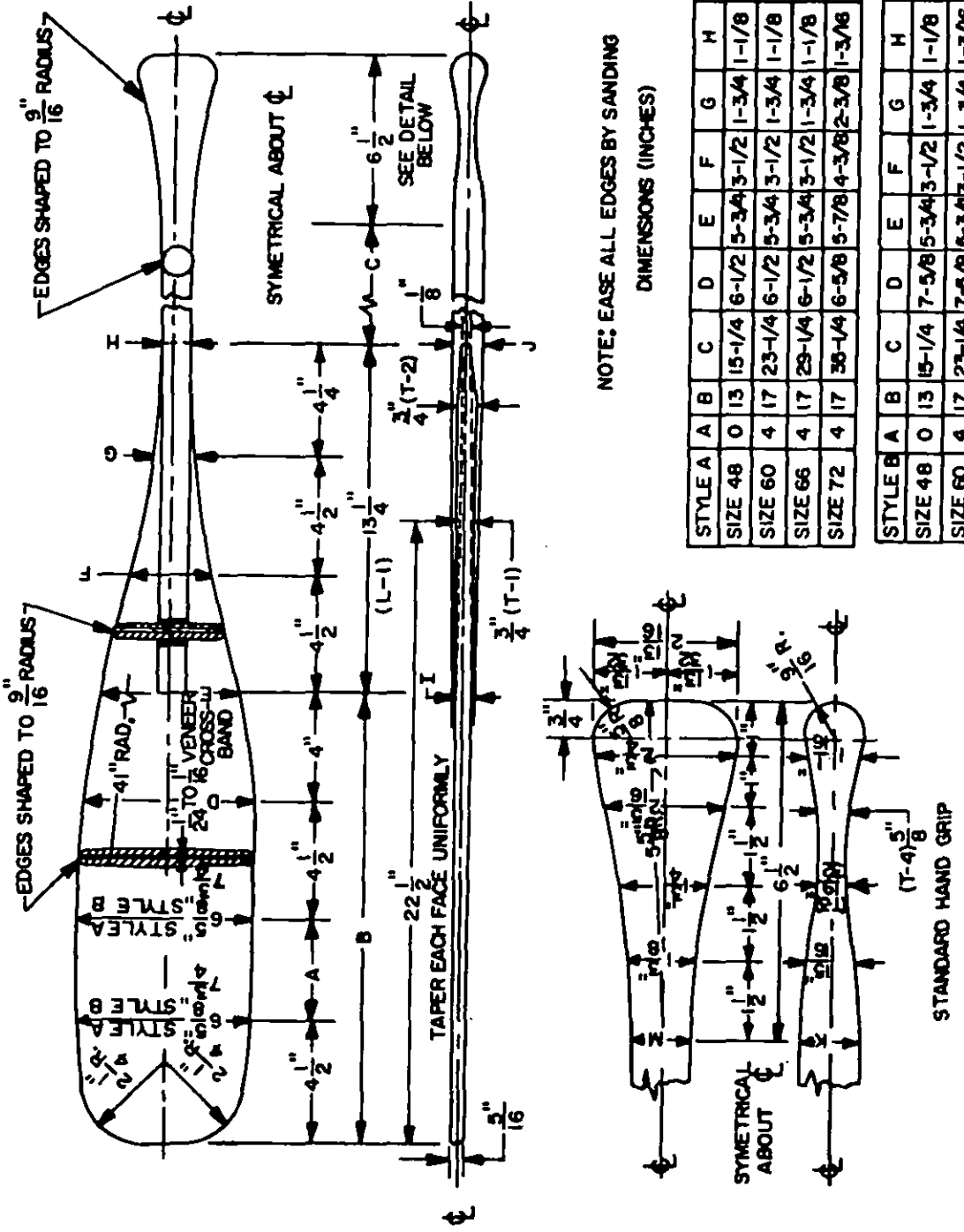


FIGURE 2. *Basic construction of type I paddle.*

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.



NOTE: EASE ALL EDGES BY SANDING

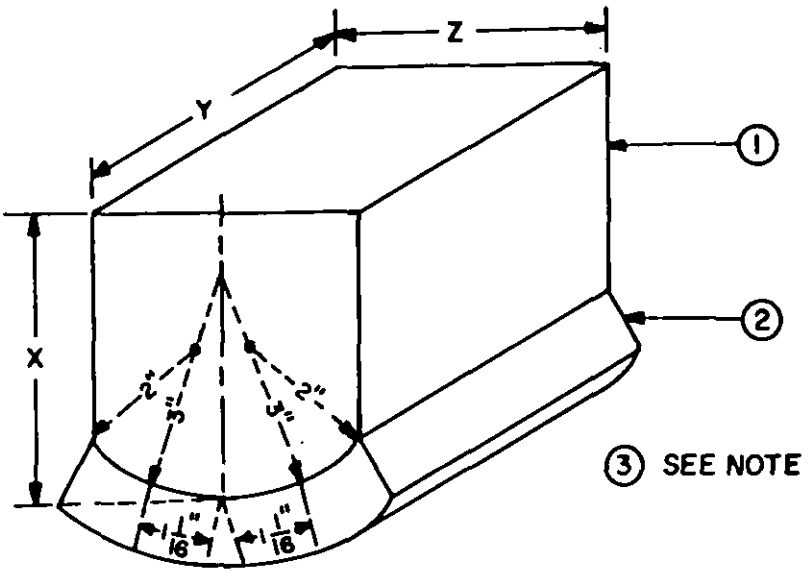
DIMENSIONS (INCHES)

STYLE A	A	B	C	D	E	F	G	H	I	J	K	M
SIZE 48	0	13	15-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	11/16	1-1/8	1-1/8	1-1/8
SIZE 60	4	17	23-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	13/16	1-1/8	1-1/8	1-1/8
SIZE 66	4	17	29-1/4	6-1/2	5-3/4	3-1/2	1-3/4	1-1/8	13/16	1-1/8	1-1/8	1-1/8
SIZE 72	4	17	35-1/4	6-5/8	5-7/8	4-3/8	2-3/8	1-3/8	13/16	1-3/16	1-3/16	1-3/16

STYLE B	A	B	C	D	E	F	G	H	I	J	K	M
SIZE 48	0	13	15-1/4	7-5/8	5-3/4	3-1/2	1-3/4	1-1/8	11/16	1-1/8	1-1/8	1-1/8
SIZE 60	4	17	23-1/4	7-5/8	5-3/4	3-1/2	1-3/4	1-3/16	13/16	1-3/8	1-3/16	1-3/16
SIZE 66	4	17	29-1/4	7-5/8	5-3/4	3-1/2	1-3/4	1-3/16	13/16	1-3/16	1-3/16	1-3/16
SIZE 72	4	17	35-1/4	7-3/4	5-7/8	4-3/8	2-3/8	1-3/16	13/16	1-3/16	1-3/16	1-3/16

FIGURE 3. *Shape and dimension of type I paddle.*

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.



1. LOADING BLOCK- MAPLE BIRCH, OAK, OR HICKORY
2. PADDING - FLEXIBLE VINYL FOAM 5-8 LB. DENSITY
3. Y-DIRECTION SHALL BE PERPENDICULAR TO THE TEST SPAN

NOTE: DIMENSIONS X, Y AND Z MAY BE VARIED TO SUIT BEND TEST APPARATUS.

FIGURE 4. *Static bending strength test loading block.*

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-15737C(SH)	29 APRIL 1991

3. DOCUMENT TITLE
PADDLES, BOAT, SOLID AND LAMINATED WOOD

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact: Mr. John Pinto (SEA 51431) PLEASE ADDERSS ALL CORRESPONDENCE AS FOLLOWS	(703) 602-0146	8-332-0146
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:54Z
Check the source to verify that this is the current version before use.